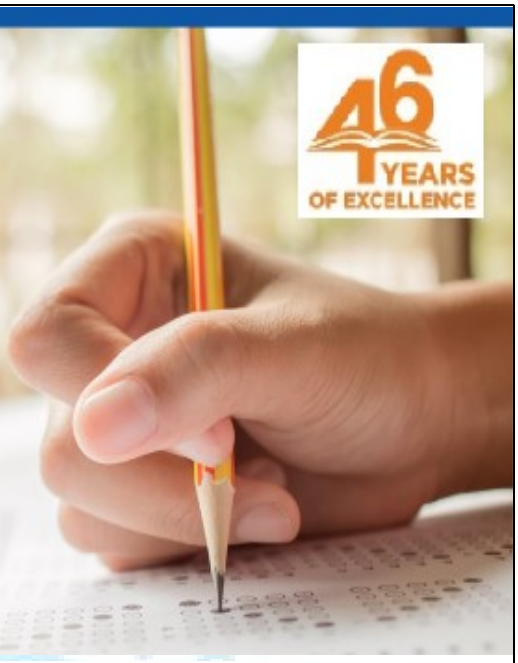
NARAYANA IIT ACADEMY INDIA

# JEE ADVANCED GRAND TEST SERIES



46 YEARS OF EXCELLENCE

NARAYANA IIT - JEE ACADEMY - INDIA

**Sec: OSR.IIT _*CO-SC(MODEL-A&APEX)** **Date: 05-04-26**
**Time: 3 Hrs** **Max. Marks: 180**

**05-04-26_OSR.IIT_*CO-SC(MODEL-A&APEX)_JEE ADV_GTA-3(P1)_SYLLABUS**

Rankers Academy JEE

**MATHEMATICS:** **TOTAL SYLLABUS**

**PHYSICS:** **TOTAL SYLLABUS**

**CHEMISTRY:** **TOTAL SYLLABUS**

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TIME: 3HRS **IMPORTANT INSTRUCTIONS** Max Marks: 180

## MATHEMATICS :

| Section | Question Type | +Ve Marks | - Ve Marks | No.of Qs | Total marks |
|---|---|---|---|---|---|
| Sec – I (Q.N : 1 – 8) | Questions with Numerical Value Type (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30) | +3 | 0 | 8 | 24 |
| Sec – II (Q.N : 9 – 14) | One of More Correct Options Type (partial marking scheme) (+1) | +4 | -2 | 6 | 24 |
| Sec – III (Q.N : 15 – 18) | Matrix Matching Type | +3 | -1 | 4 | 12 |
| **Total** | | | | **18** | **60** |

## PHYSICS:

| Section | Question Type | +Ve Marks | - Ve Marks | No.of Qs | Total marks |
|---|---|---|---|---|---|
| Sec – I (Q.N : 19 – 26) | Questions with Numerical Value Type (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30) | +3 | 0 | 8 | 24 |
| Sec – II (Q.N : 27 – 32) | One of More Correct Options Type (partial marking scheme) (+1) | +4 | -2 | 6 | 24 |
| Sec – III (Q.N : 33 – 36) | Matrix Matching Type | +3 | -1 | 4 | 12 |
| **Total** | | | | **18** | **60** |

## CHEMISTRY:

| Section | Question Type | +Ve Marks | - Ve Marks | No.of Qs | Total marks |
|---|---|---|---|---|---|
| Sec – I (Q.N : 37 – 44) | Questions with Numerical Value Type (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30) | +3 | 0 | 8 | 24 |
| Sec – II (Q.N : 45 – 50) | One of More Correct Options Type (partial marking scheme) (+1) | +4 | -2 | 6 | 24 |
| Sec – III (Q.N : 51 – 54) | Matrix Matching Type | +3 | -1 | 4 | 12 |
| **Total** | | | | **18** | **60** |

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**MATHEMATICS** **Max. Marks: 60**

## SECTION - I
## (Maximum Marks : 24)

This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**
For each question, enter the correct numerical value (in decimal notation, truncated/rounded off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30) designated to enter the answer.
Answer to each question will be evaluated according to the following marking scheme:
**Full Marks:** +3 If ONLY the correct numerical value is entered as answer.
**Zero Marks:** 0 In all other cases.

1. Consider a function $f$ defined on the set of non-negative integers such that $f(0)=1,\ f(1)=0$ and $f(n)+f(n-1)=n\ f(n-1)+(n-1)f(n-2)$ for $n\geq 2$ then $f(6)$ equals

2. Let $a$, $b$ and $c$ three non-negative numbers such that

$a^3-a^2+a=b$

$b^3-b^2+b=c$

$c^3-c^2+c=a$

Number of possible triplets of type $(a, b, c)$ is/are

3. Let $p$ & $q$ are real numbers such that

$(\tan p-1)^3+2026(\tan p-1)=-1 \& (1-\cot q)^3+2026(1-\cot q)=-1$

$\tan p\neq \cot q$, then number of possible values of $r$ which satisfy the equation $\tan p+\cot q+\sin r+\cos r=3$ in $x\in[-2\pi, 2\pi]$ is

4. Let $x$ and $y$ be two real numbers such that $2\sin x\sin y+3\cos y+6\cos x\sin y=7$, then $\tan^2 x+\tan^2 y$ is equal to _____.

5. $\lim\limits_{x\to 0}\dfrac{x^2\sin\beta x}{\tan\alpha x-\sin x}=4$ where $\alpha, \beta\in R,$ then find the value of $(\alpha+2\beta)$

6. Let $f(x)=\begin{cases}(x^3+a)\sin(x+b), & x<0\\ \tan^{-1}x^4+x^2-x+1, & x\geq 0\end{cases}$ is differentiable function for all real $x$ and $a\in R, b\in[0, 10\pi]$, then number of ordered pair of $(a, b)$ is-

7. Let $a, b$ are two integers such that $0<a<b<10^6$ and arithmetic mean of $a$ and $b$ is exactly 8 more than its geometric mean. If number of such ordered pairs is $N$, then $N$ is equal to

8. Let $z_1$ and $z_2$ are two distinct roots of the equation $z^{101}=1$ then the probability that $|z_1+z_2|\geq\sqrt{2+\sqrt{3}}$ is



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## SECTION – II

**(Maximum Marks : 24)**

This section contains **SIX(06)** multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which **ONE OR MORE THAN ONE** option can be correct.

**Marking scheme: +4 for all correct options & +1 partial marks, 0 if not attempted and -2 in all wrong cases**

9. Let $H: y(3y+4x)=-4$ is a hyperbola and $y=mx+c$ is its conjugate axis. Length of latus rectum of $H$ is $L$, eccentricity $e$ and $(x_1, y_1)$ is one vertex with $y_1>0$, then-

A) $m=3$ B) $L=1$ C) $4e^2=5$ D) $5y_1^2=4$

10. If $P(x)=mid\,(g(x), h(x), f(x))$ means the function will be second in order when values of the three functions are arranged at a particular $x$.

If $P(x)=mid\left(x-1, (x-3)^2, 3-\frac{(x-3)^2}{2}\right), x\in[1, 3]$ then for given interval:

A) Number of points of discontinuity of $P(x)$ will be 0

B) Number of points of discontinuity of $P(x)$ will be 1

C) Number of points of non-derivability of $P(x)$ will be 1

D) Number of points of non-derivability of $P(x)$ will be 2

11. Let $P(x)$ be a polynomial of least degree with leading coefficient 1 and rational coefficients, such that $P\left(\sqrt{5+\sqrt{5+\sqrt{5+.....\infty \text{ terms}}}}\right)=0$ then:

A) The value of $100\sum_{r=2}^{100}\frac{1}{P(r)+5}$ is equal to 99

B) $\lim_{n\to\infty}\sum_{r=3}^{n}\frac{1}{P(r)+3}$ is equal to 1

C) $\int_0^1\frac{dx}{P(x)+x+6}$ is equal to $\frac{\pi}{4}$

D) $\int_1^2\frac{dx}{P(x)+x+4}$ is equal to $\ln\left(2+\sqrt{3}\right)$



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12. For $x \in \left(0, \frac{\pi}{4}\right)$

$$R_n = \sum_{r=1}^{2n} \sin\left(\sin^{-1} x^{4r-3}\right), \quad S_n = \sum_{r=1}^{2n} \cos\left(\cos^{-1} x^{4r-2}\right),$$

$$T_n = \sum_{r=1}^{2n} \tan\left(\tan^{-1} x^{4r-1}\right), \quad U_n = \sum_{r=1}^{2n} \cot\left(\cot^{-1} x^{4r}\right)$$

Where $n \in N$ and $n \geq 4$

A) $R_n < S_n < T_n < U_n$

B) $R_n > S_n > T_n > U_n$

C) $\lim_{n\to\infty}\left(R_n + S_n + T_n + U_n\right)$ is equal to $\frac{x}{1-x}$

D) The value of $x$ for which $R_n + S_n = T_n + U_n$ is $2\sin\frac{\pi}{10}$

13. Let $P$ be the plane consisting of all point that are equidistant from $A(-4, 2, 1)$ and $B(2, -4, 3)$ and Q denotes the plane $x - y + Cz = 1$, where $C \in R$.

A) If $C = \frac{1}{3}$ then plane $P$ will be parallel to $Q$

B) If $C = -1$ then plane $P$ will be perpendicular to $Q$

C) Volume of tetrahedron formed by plane P with coordinate planes will be $\frac{2}{81}$.

D) If line $L: \frac{x-1}{1} = \frac{y+2}{3} = \frac{z-7}{-1}$ intersect plane $P$ at $R(\alpha, \beta, \gamma)$, then $\alpha + \beta + \gamma = 12$

14. Let $A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix}$, then the correct statements is:

A) $A^2 - 4A - 5I_3 = 0$

B) $A^{-1} = \frac{1}{20}\left(A^2 - 21I_3\right)$

C) $A^3 = 3A^2 + 9A + 5I_3$

D) $A^{-1} = \frac{1}{5}\left(2A - 5I_3\right)$



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## SECTION - III
## (Maximum Marks: 12)

This section contains **FOUR (04)** questions.
Each question has **TWO (02)** matching lists: **LIST-I** and **LIST-II**.
**FOUR** options are given representing matching of elements from **LIST-I** and **LIST-II**. **ONLY ONE** of these four options corresponds to a correct matching.
For each question, choose the option corresponding to the correct matching.
For each question, choose the option corresponding to the correct matching.
**Full Marks** : +3 If ONLY the option corresponding to the correct matching is chosen.
**Zero Marks** : 0 If none of the options is chosen (i.e. the question is unanswered).
**Negative Marks** : –1 In all other cases.

15. Match the following.

| | **Column-I** | | **Column-II** |
|---|---|---|---|
| I) | Let $S_1$ and $S_2$ are two point on $AB$ of a $\Delta ABC$ with vertices $A(-2,3)$, $B(4,-6)$ and $C(1,1)$. $CS_1$ and $CS_2$ divide the triangle into three parts of equal area. If the equation of the lines through the origin drawn parallel to $CS_1$ and $CS_2$ is $y^2+\lambda xy-\mu x^2=0$ then the value of $(\lambda+\mu)$, is | P) | 2 |
| II) | If the area enclosed by the curves $y=\lvert x\rvert$ and $x^2+y-2=0$ is $\frac{p}{q}$, where $p$ and $q$ are coprime then the value of $(p-q)$, is | Q) | 4 |
| III) | The curve that satisfies the differential equation $\frac{dy}{dx}=\frac{x^2+y^2}{2xy}$ is a hyperbola passing through $(2,1)$ with eccentricity $e$. The values of $e^4$ is equal to | R) | 6 |
| IV) | If $\lim_{n\to\infty}\frac{1}{n^{2m}}\left[\left(n^2+1^2\right)^m\left(n^2+2^2\right)^m.....\left(n^2+n^2\right)^m\right]^{\frac{1}{n}}=\left(\frac{a\sqrt{e^{\pi}}}{e^2}\right)^m$, then the value of $a$, is | S) | 7 |
| | | T) | 8 |

A) I-T; II-Q; III-Q; IV-R
B) I-T; II-Q; III-S; IV-P
C) I-S; II-Q; III-Q; IV-P
D) I-S; II-R; III-Q; IV-R


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16. Match the following.

| | Column-I | | Column-II |
|---|---|---|---|
| I) | If the distance of the plane passing through the point $P(1, 1, 1)$ and perpendicular to the line $\frac{x-1}{3}=\frac{y-1}{0}=\frac{z-1}{4}$ from the origin is $\frac{p}{q}$ (where $p$ and $q$ are coprime), then the value of $(p-q)$ is | P) | 1 |
| II) | The interval $[0, 4]$ is divided into $n$ equal sub-intervals by the points $x_0, x_1, x_2....,x_{n-1}, x_n$ where $0=x_0<x_1<x_2<x_3.....<x_n=4$. If $\delta x=x_i-x_{i-1}$ for $i=1,2,3,.....n$ then the value of $\lim\limits_{\delta x\to 0}\sum\limits_{i=1}^{n}x_i\delta x$ is | Q) | 2 |
| III) | Let $f:R\to R$ be a function satisfying $xf(x)+(1-x)f(-x)=x^2+x+1$ for any real number $x$. The greatest real number $M$ for which $f(x)\geq M$ for all real numbers $x$, is equal to $\frac{p}{q}$, where $p$ and $q$ are coprime. The value of $(q-p)$, is | R) | 3 |
| IV) | The area of the trapezium $ABCD$ with $AB \parallel CD$, $AD\perp AB$ and $AB=3CD$ is equal to 4. A circle inside the trapezium is tangent to all of its sides. If the radius of the circle is $r$ then the value of $4r^2$, is | S) | 6 |
| | | T) | 8 |

A) I-R; II-T; III-P; IV-Q B) I-Q; II-P; III-T; IV-R

C) I-R; II-T; III-S; IV-Q D) I-Q; II-T; III-P; IV-R

17. Match the following.

| | Column-I | | Column-II |
|---|---|---|---|
| I) | Total number of values of $x$, of the form $\frac{1}{n}$, $n\in N$ in the interval $\left[\frac{1}{15},\frac{1}{10}\right]$ which satisfies $\{x\}+\{2x\}+.....+\{12x\}=78x$, (where $\{\ \}$ denotes fractional part of $x$) | P) | 1 |



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| | | | |
|---|---|---|---|
| II) | Let $\left(2x^2+3x+4\right)^{10}=\sum_{r=0}^{20}a_r x^r$ then the value of $\frac{a_9}{a_{11}}$ is equal to | Q) | 2 |
| III) | Let $G$ be centroid of $\Delta ABC$ whose side length are $a, b, c$. If $P$ is a point in the plane of $\Delta ABC$ such that $PA=1, PB=2, PC=1$ & $PG=1$ then the value of $\frac{a^2+b^2+c^2}{9}$ is equal to | R) | 3 |
| IV) | If a derivable function $f:R^+\to R$ satisfies $f(xy)=f(x)+f(y)\forall x,y\in R^+$. If $f(16)=3$ then $\frac{1}{f(2)}+\frac{1}{f(4)}$ is equal to | S) | 4 |
| | | T) | 10 |

A) I-S; II-Q; III-P; IV-P
B) I-R; II-S; III-P; IV-P
C) I-S; II-S; III-P; IV-Q
D) I-R; II-Q; III-P; IV-Q

18. Match the following.

| | **Column-I** | | **Column-II** |
|---|---|---|---|
| I) | The remainder when $25^{24^{26}}$ is divided by 7 is | P) | 0 |
| II) | $x^8+y^8+6=8xy$, where $x,y\in I$ Number of ordered pair $(x, y)$ is | Q) | 1 |
| III) | If $y=3x-8$ is tangent at $(7, 13)$ on a parabola with focus $(-1, -1)$ and length of latus rectum is $l$, then $[l]$ is equal to | R) | 2 |
| IV) | A hyperbola has centre at origin and one focus at $(6, 8)$. If its two directrices are $3x+4y+10=0$ and $3x+4y-10=0$ and eccentricity is $e$, then the value of $\frac{4e^2}{5}$ is equal to | S) | 3 |
| | | T) | 4 |

[**Note:** $[k]$ denotes greatest integer less than or equal to $k$.]

A) I-Q; II-R; III-R; IV-T
B) I-T; II-R; III-R; IV-S
C) I-T; II-S; III-R; IV-T
D) I-Q; II-S; III-Q; IV-S

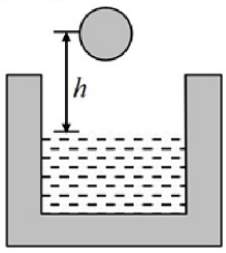

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**PHYSICS** **Max. Marks: 60**

## SECTION - I
## (Maximum Marks : 24)

This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**
For each question, enter the correct numerical value (in decimal notation, truncated/rounded off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30) designated to enter the answer.
Answer to each question will be evaluated according to the following marking scheme:
**Full Marks:** +3 If ONLY the correct numerical value is entered as answer.
**Zero Marks:** 0 In all other cases.

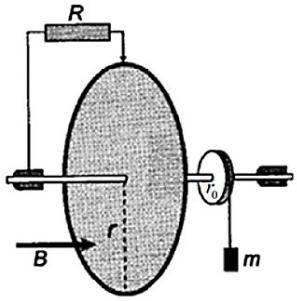
19. A small ball of radius $r$ and density $\rho$ falls freely under gravity through a distance $h$ before entering water. Velocity of ball does not change even on entering water (density $\sigma$). If viscosity of water is $\eta$, then the value of $h$ is given by $\frac{gr^4}{n}\left(\frac{\rho-\sigma}{\eta}\right)^2$. Find the value of $n$.




20. Moseley's law for $K_\alpha$ photon is given by $\sqrt{v}=a(Z-b)$ where $a$ is a universal constant and $b$ is a screening constant. Moseley's logic helps us assume $b=1$ for $K_\alpha$ photon. If there is a percentage error $=10^{-2}$ in the measurement of $b$ due to actual electronic orbital configuration of an atom, then the relative error in the measurement of $v$ if $Z=51$ is $c\times10^{-d}$ in scientific notation. The value of $c+d$ is _____.

21. A radioactive source is a mixture of two radioactive materials X and Y with mean life-time of 6 days and 9 days respectively. Initially $20\%$ of radiations came from the radioactive decay of X and remaining from Y. The time in past at which $80\%$ of radiations came from the decay of X is _____ days. $[\ln 2=0.30]$

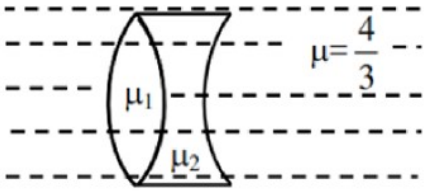
22. A conducting disc of radius $r$ can rotate about a horizontal conducting shaft. A smaller non conducting disc of radius $r_0$ is fixed onto the same shaft and has an ideal mass less cord wrapped around it, which is attached to a small object of mass $m$ hanging vertically as shown. Two ends of the resistor $R$ are connected to the perimeter of the disc and to the shaft by sliding contacts. The system is then placed into a uniform horizontal magnetic field $B$ and the mass is released. If $r=10\ cm, r_0=2\ cm, R=0.01\ \Omega, B=0.2T, m=50\ g$ then find the terminal angular velocity (in $\text{rad/s}$) of the disk. $[g=9.8\ ms^{-2}]$



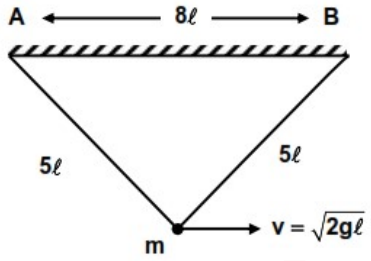



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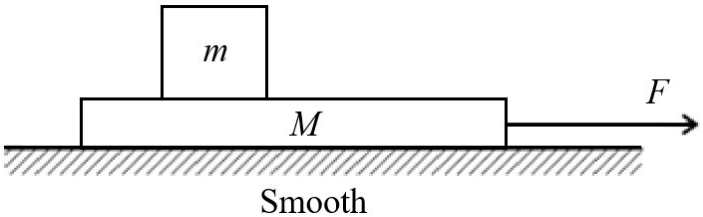
23. Two thin symmetrical lens of different nature have equal radii of curvature of all faces $R = 20\ cm$. The lenses are put close together and immersed in water. Refractive index of water is $\frac{4}{3}$. The focal length of the system is 24 *cm*. The difference between refractive indices $(\mu_1 - \mu_2)$ of the two lenses is _____.




24. An ideal smooth massless inextensible string whose ends are fixed at A and B passes through the bead as shown. Initially the system is in equilibrium in a vertical plane. Now bead is given a velocity $\sqrt{2gl}$ in horizontal direction and in the plane of the string as shown. The tension in the string just after the velocity is given is $\frac{nmg}{3}$. Find the value of *n*.




25. A block of mass $m = 6\ kg$ sits on a plank of mass $M = 10\ kg$ which lies on a smooth surface. The coefficient of friction between the block and the plank is $\mu = 0.5$. At $t = 0$, a time dependent force $F = 2t\ N$ starts to act on the plank ($t$ is in seconds). Find the time (in $s$) after which the block starts to slip on the plank? [$g = 10\ ms^{-2}$]




26. Two factories are sounding their sirens at 600 *Hz* each. A man goes from one factory to the other at a speed of 2 $ms^{-1}$. The velocity of sound in air is 320 $ms^{-1}$. Due to this how many beats he hears per minute?

## SECTION – II
## (Maximum Marks : 24)

This section contains **SIX(06)** multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which **ONE OR MORE THAN ONE** option can be correct.

**Marking scheme: +4 for all correct options & +1 partial marks, 0 if not attempted and -2 in all wrong cases**

27. A particle moves on a plane from an initial point *A* to a final point *B* along some smooth path. Select the incorrect option(s) for this motion.

A) The velocity of the particle is always along $\overrightarrow{AB}$

B) The velocity of the particle can never be along $\overrightarrow{AB}$

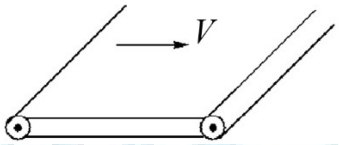


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C) The velocity of the particle should be along $\overrightarrow{AB}$ at least once during its motion

D) The velocity of the particle should be along $\overrightarrow{AB}$ at least twice during its motion

28. Charge is deposited at a constant rate by the left roller onto the large non conducting belt. The belt gets a uniform surface charge density $\sigma$, as it moves with a speed $V$ between the rollers as shown. The charge is removed by the right roller. For a point just above the belt mark the correct option(s).



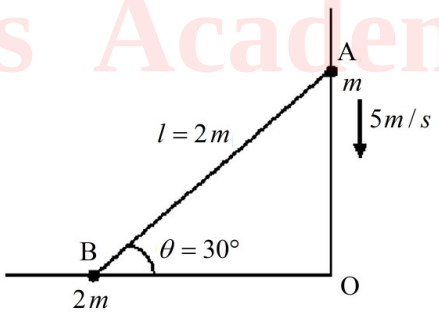

A) Magnetic field is $\frac{\mu_0 \sigma V}{2}$

B) Magnetic field is $\mu_0 \sigma V$

C) If the roller spin twice as fast then the magnetic field becomes $\mu_0 \sigma V$

D) If the rollers spin twice as fast then the magnetic field becomes $\frac{\mu_0 \sigma V}{4}$

29. Two particles A and B of mass $m$ and $2m$ are connected by a light rigid rod of length $l$. The particles are free to slide on frictionless $L$-shaped horizontal frame as shown in the figure. A moves with a constant velocity of $5\ m \cdot s^{-1}$. Then at the instant shown




A) Speed of B is $\frac{10}{\sqrt{3}} m \cdot s^{-1}$

B) The rate of change of $\theta$ is $\frac{5}{\sqrt{3}} rad \cdot s^{-1}$

C) Speed of B is $\frac{5}{\sqrt{3}} m \cdot s^{-1}$

D) The rate of change of $\theta$ is $\frac{-10}{\sqrt{3}} rad \cdot s^{-1}$

30. A small body of mass $m$ is projected from the surface of a spherical planet of mass $M$ and radius $R$ with escape velocity in the radially outward direction. Assuming that only gravitational force due to the planet acts on the body during its motion, which of the following options are **CORRECT?**

A) Speed of the body when it is at height of $2R$ from the surface of planet is $\sqrt{\frac{2GM}{3R}}$

B) Speed of the body when it is at height of $R$ from the surface of planet is $\sqrt{\frac{GM}{R}}$



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C) Time taken by the body to reach at a height of $3R$ from the surface of planet is $\frac{7}{3}\sqrt{\frac{2R^3}{GM}}$

D) Speed of the body when it is at height of $R$ from the surface of planet is $\sqrt{\frac{GM}{2R}}$

31. A thin spherical quarter shell has a uniform surface charge density $\sigma$ on its outer surface. It's radius is $R$.

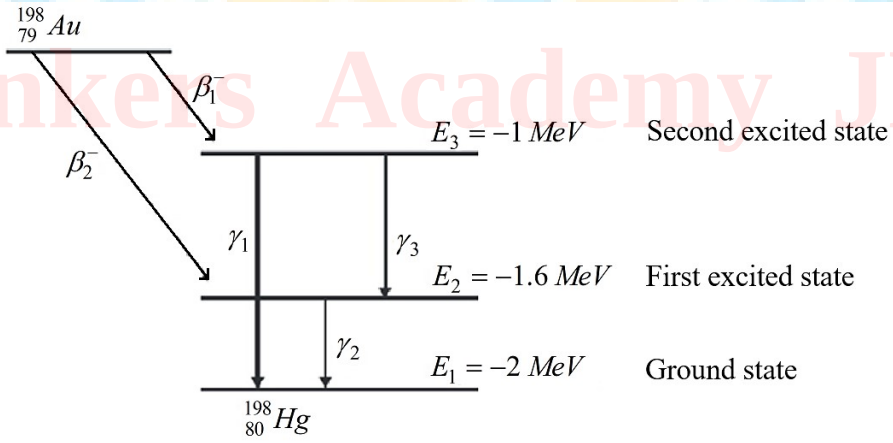
A) The potential at the centre of curvature is $\frac{\sigma R}{4\varepsilon_0}$

B) The potential at the corner of the shell is $\frac{\sigma R}{4\varepsilon_0}$

C) Magnitude of electric field at the centre of curvature is $\frac{\sigma}{4\sqrt{2}\varepsilon_0}$

D) Magnitude of electric field at the centre of curvature is $\frac{\sigma}{2\sqrt{2}\varepsilon_0}$

32. Gold Nucleus $^{198}_{79}Au$ can decay into mercury nucleus $^{198}_{80}Hg$ by two decay schemes shown in figure. The energy levels of the nucleus are shown in figure.




(i) It can emit a $\beta$ particle ( $\beta_1$ ) and come to ground state by either emitting one $\gamma$ ray $(\gamma_1)$ or emitting two $\gamma$ rays $(\gamma_2$ and $\gamma_3)$ and come to ground state

(ii) It can emit a $\beta$ particle ( $\beta_2$ ) and come to ground state by either emitting one $\gamma$ ray $(\gamma_2)$

Atomic masses: $^{198}_{79}Au\,197.9682\,amu; ^{198}_{80}Hg\,197.9662\,amu; 1\,amu = 930\ MeV\cdot c^{-2}$

A) The maximum kinetic energy of emitted $\beta_2$ particles is 1.46 MeV

B) The maximum kinetic energy of emitted $\beta_2$ particles is 1.86 MeV

C) The maximum kinetic energy of emitted $\beta_1$ particles is 0.86 MeV

D) The maximum kinetic energy of emitted $\beta_1$ particles is 1.28 MeV

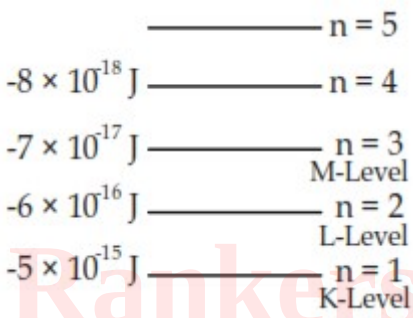


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## SECTION - III
## (Maximum Marks: 12)

This section contains **FOUR (04)** questions.
Each question has **TWO (02)** matching lists: **LIST-I** and **LIST-II**.
**FOUR** options are given representing matching of elements from **LIST-I** and **LIST-II**. **ONLY ONE** of these four options corresponds to a correct matching.
For each question, choose the option corresponding to the correct matching.
For each question, choose the option corresponding to the correct matching.
**Full Marks** : +3 If ONLY the option corresponding to the correct matching is chosen.
**Zero Marks** : 0 If none of the options is chosen (i.e. the question is unanswered).
**Negative Marks** : –1 In all other cases.

33. Energy levels of a hypothetical atom are shown in the figure. The hypothetical atom is bombarded with high energetic electrons. The bombarded electrons have ability to complete removal of electrons from K – level, this vacancy is filled by de-excitation of electrons from L -level to K – level, this energy may appear as X – rays or may all be used to kick out electron from M – level of the atoms. Now match the List – I with List – II.




| List - I | | List - II | |
|---|---|---|---|
| P) | The minimum potential difference (in volt) of high energetic electron required to kick out the electrons from K – level of atom. | 1) | $4.33\times10^{-15}$ |
| Q) | Energy released (in Joule) when L – level electron moves to fill the vacancy in the K – level. | 2) | $4.4\times10^{-15}$ |
| R) | Wave length of X – ray emitted (in joule) in meter | 3) | $5.12\times10^{-15}$ |
| S) | Kinetic energy (in joule) of electron emitted from the M – level | 4) | $3.125\times10^{4}$ |
| | | 5) | $4.54\times10^{-11}$ |

The correct option is:

A) $(P)\rightarrow(4);(Q)\rightarrow(2);(R)\rightarrow(5);(S)\rightarrow(1)$

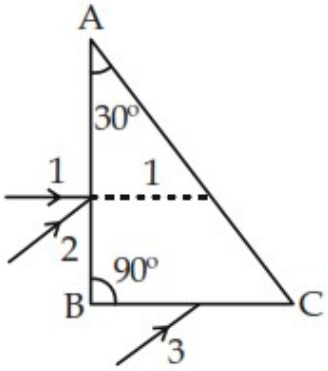
B) $(P)\rightarrow(4);(Q)\rightarrow(2);(R)\rightarrow(5);(S)\rightarrow(3)$

C) $(P)\rightarrow(4);(Q)\rightarrow(3);(R)\rightarrow(2);(S)\rightarrow(1)$

D) $(P)\rightarrow(4);(Q)\rightarrow(2);(R)\rightarrow(3);(S)\rightarrow(1)$


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34. ABC is a right-angled prism kept in air. A ray (1) is incident on the face AB along the normal. Refractive index of the material of prism is the minimum value that will be required so that ray (1) undergoes total internal reflection at the face AC. Another ray (2) is incident on the face AB such that it emerges from face AC along the normal to AC. A third ray (3) falls on the face BC and emerges from face AC such that its angle of emergence is the same as that of incidence. Assuming light (1), (2) and (3) have the same wavelength, then match the following.




| List - I | | List - II | |
|---|---|---|---|
| P) | Refractive index of the material of prism is | 1) | 120 |
| Q) | Angle of incidence in degree of ray (2) is | 2) | 90 |
| R) | Deviation in degree suffered by ray (2) is | 3) | 2 |
| S) | Deviation in degree suffered by ray (3) is | 4) | 60 |
| | | 5) | 1.5 |

A) P – 3, Q – 2, R – 1, S – 2 B) P – 3, Q – 2, R – 4, S – 1

C) P – 4, Q – 1, R – 2, S – 3 D) P – 3, Q – 1, R – 4, S – 2

35. The speed $(v)$ of a particle of mass 1 $kg$ moving along a straight line, when it is at a distance $(x)$ from a fixed point on the line, is given by, $v^2 = 144 - 9x^2$. Match the following.

| List - I | | List - II | |
|---|---|---|---|
| P) | Motion is simple harmonic with time period | 1) | $\frac{2\pi}{3}$ units |
| Q) | Maximum displacement from the fixed point is | 2) | 27 units |



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| | | | |
|---|---|---|---|
| R) | Magnitude of acceleration at a distance 3 units from the fixed point is | 3) | 4 units |
| S) | Potential energy at a distance $x = 2$ unit from fixed point is | 4) | 2 units |
| | | 5) | 18 units |

A) $P-1, Q-4, R-2, S-3$ B) $P-1, Q-3, R-5, S-2$

C) $P-3, Q-1, R-2, S-5$ D) $P-1, Q-3, R-2, S-5$

36. A vertical solenoid of length 120 *cm* and diameter 2.50 *cm* has 1400 turns of copper wire carrying a current of 2.00 *A* in the counter clockwise direction as seen from the above. A superconducting bar of diameter 2.20 *cm* is inserted such that its lower end lies deep inside the solenoid while the upper end is outside. The bar exhibits perfect diamagnetism.

Match the quantities in List-I with their corresponding values or directions in List-II.

| List - I | | List - II | |
|---|---|---|---|
| P) | Magnetic field inside the solenoid (in *mT*) | 1) | 2.93 |
| Q) | Energy density of the magnetic field (in *Pa*) | 2) | 3.42 |
| R) | Magnitude of force on the superconducting bar (in *mN*) | 3) | 1.30 |
| S) | Direction of surface current on the bar (as viewed from above) | 4) | Clockwise |
| | | 5) | Counterclockwise |

A) $P \to 1, Q \to 2, R \to 3, S \to 4$ B) $P \to 2, Q \to 1, R \to 4, S \to 5$

C) $P \to 1, Q \to 2, R \to 3, S \to 5$ D) $P \to 3, Q \to 1, R \to 2, S \to 4$



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**CHEMISTRY** **Max. Marks: 60**

## SECTION - I
## (Maximum Marks : 24)

This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**
For each question, enter the correct numerical value (in decimal notation, truncated/rounded off to the **second decimal place;** e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30) designated to enter the answer.
Answer to each question will be evaluated according to the following marking scheme:
**Full Marks:** +3 If ONLY the correct numerical value is entered as answer.
**Zero Marks:** 0 In all other cases.

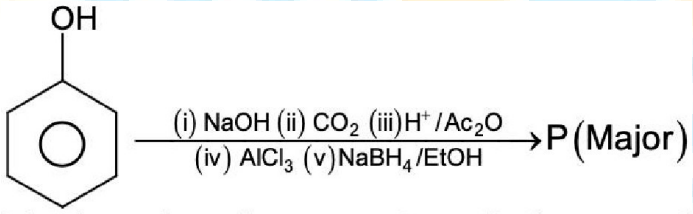
37. The overall formation constant of $\left[Co(NH_3)_6\right]^{2+}$ in aqueous solution is $10^5$ and the standard potential for reduction of $Co^{3+}(aq)$ and $\left[Co(NH_3)_6\right]^{3+}(aq)$ are as follows

$Co^{3+}(aq)+e \rightleftharpoons Co^{2+}(aq)\ E^\circ = 1.90$

$\left[Co(NH_3)_6\right]^{3+}(aq)+e \rightleftharpoons \left[Co(NH_3)_6\right]^{2+}(aq);\quad E^\circ = +0.1V$

It the nearest overall formation constant of the $\left[Co(NH_3)_6\right]^{3+}$ ion is $10^x$. Then the value of $x$ would be (Given: $E_{cell} = E^\circ_{cell} - 0.06\log Q$]

38. The activation energy for a reaction is $30.8\times298\times4.606$ cal/mol. If the increase in the rate constant when its temperature is increased from 298 $K$ to 308 $K$ is $x$ % then '$x$' would be

39. 


Total number of oxygen atoms in the aromatic product (P) is

40. If time required when a current of 9.65 amp is passed through 80 ml of 0.1 $M$ $NaCl$ solution in order to make its $pH$ = 12 is $x$ sec. and time required when a current of 965 amp is passed through an aqueous solution of $CuSO_4$ to produce 0.28 litres of oxygen gas at STP is $y$ sec. then the value of $\frac{x}{y}$ would be

41. The $pH$ of a solution formed by mixing 10 $ml$ 0.1 $M$ $NaH_2PO_4$ and 20 $ml$ $Na_2HPO_4$ would be (Given $pK_1 = 2.2,\ pK_2 = 7.2,\ \log 2 = 0.3$)

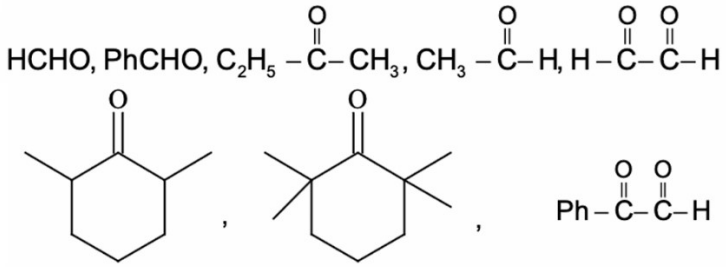


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42. Among the following compounds

(I) The compound that will not form enamine with $R_2NH$ is '$a$'.

(II) The compound which involve in Cannizzaro reaction (excluding intermolecular) is '$b$'.




Then what would be the value of $a+b$?

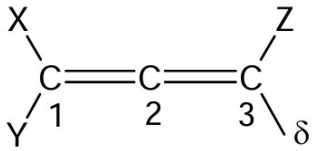
43. 6 mol of a mixture containing same number of moles of $NaNO_3$, $NH_4NO_3$, $LiNO_3$, $Ca(NO_3)_2$, $Pb(NO_3)_2$ and $AgNO_3$, was heated strongly. Total number of moles of $O_2(g)$ evolved is:

44. In a crystalline solid, atoms of $X$ forms FCC packing and the atoms of $Y$ occupy all octahedral voids. If all the atoms along one body diagonal are removed, then the simplest formula of the crystalline solid will be $X_aY_b$, the value of $(a/b)$ is:

## SECTION – II
## (Maximum Marks : 24)

This section contains **SIX(06)** multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which **ONE OR MORE THAN ONE** option can be correct.

**Marking scheme: +4 for all correct options & +1 partial marks, 0 if not attempted and -2 in all wrong cases**

45. 


The $\pi$ electron cloud of $C_1-C_2$ is present in the plane of paper, then, which of the following is incorrect?

A) $Z$ is perpendicular to the plane of paper.

B) $X$ is present in the plane of paper.

C) $\sigma$ bond of $C_2-C_3$ is perpendicular to the plane of paper.

D) $\pi$ electron cloud of $C_2-C_3$ bond and $X$ is present in same plane.

46. Inorganic benzene surpasses organic benzene in which property:

A) melting point

B) edge length of its hexagon

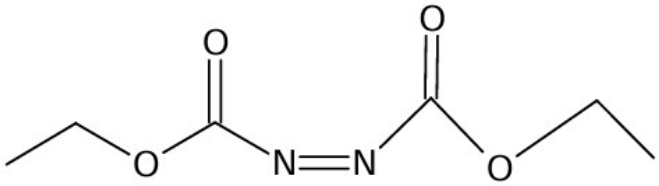
C) molar mass

D) total number of electrons in the molecule



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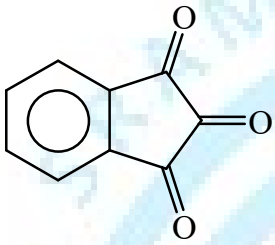
47. DEAD is an organic molecule standing for Di Ethyl Azo Dicarboxylate having the following structure:



Which of the following gas(es) will be evolved when DEAD is strongly heated with excess conc. $H_2SO_4$

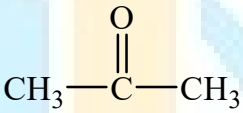
A) $N_2$ B) $C_2H_4$ C) $CO$ D) $CO_2$

48. Which of the following carbonyl compound(s) form (s) readily isolable hydrate?

A) 

B) $CCl_3 - CHO$

C) 

D) 

49. When $CO_2$ is passed into aqueous

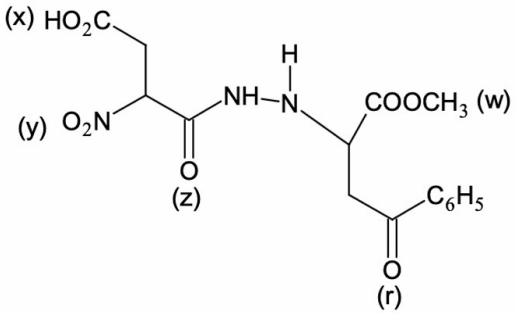
A) $Na_2CrO_4$ solution, its yellow colour changes to green

B) $K_2MnO_4$ solution, it disproportionates to $KMnO_4$ and $MnO_2$

C) $Na_2Cr_2O_7$ solution, its orange colour changes to green

D) $KMnO_4$ solution, its pink colour changes to colourless solution

50. Identify the correct statement(s) given for the reduction of the following compounds:



A) $LiAlH_4$ reduces $x, y, z, w, r$

B) $NaBH_4$ reduces '$r$' only

C) $Na / C_2H_5OH$ reduces $x$, $r$ and $w$ only

D) $(CH_3)_2CHOH + \left[(CH_3)_2CHO\right]_3 Al, \Delta$ reduces only '$r$'

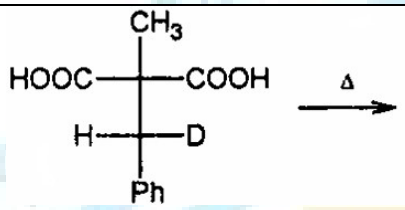
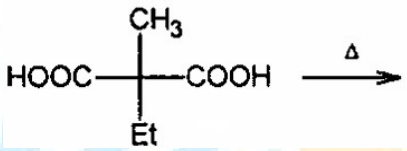
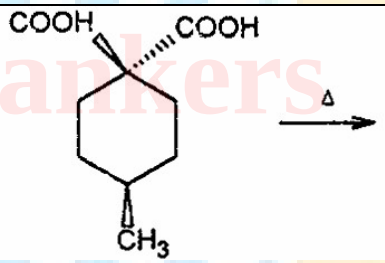
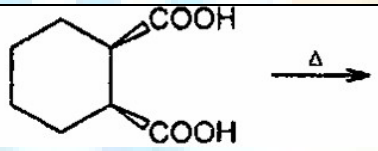


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## SECTION - III
## (Maximum Marks: 12)

This section contains **FOUR (04)** questions.
Each question has **TWO (02)** matching lists: **LIST-I** and **LIST-II**.
**FOUR** options are given representing matching of elements from **LIST-I** and **LIST-II**. **ONLY ONE** of these four options corresponds to a correct matching.
For each question, choose the option corresponding to the correct matching.
For each question, choose the option corresponding to the correct matching.
**Full Marks** : +3 If ONLY the option corresponding to the correct matching is chosen.
**Zero Marks** : 0 If none of the options is chosen (i.e. the question is unanswered).
**Negative Marks** : –1 In all other cases.

51. Match the following:

| | Column-I | | Column-II |
|---|---|---|---|
| I) |  | P) | Diastereomer |
| II) |  | Q) | Racemic mixture |
| III) |  | R) | Meso compound |
| IV) |  | S) | $CO_2$ gaswill evolve |
| | | T) | Dehydration will occur |

A) I-Q,S; II-P,S; III-P,S; IV-R,T
B) I-P,T; II-Q,T; III-P,S; IV-R,T
C) I-P,S; II-Q,S; III-P,S; IV-R,T
D) I-P,S; II-Q,S; III-P,T; IV-Q,T

52. Match the following Column, Column I with Column II:

| | Column-I | | Column-II |
|---|---|---|---|
| I) | $[Ma_3b_2c]$ | P) | All stereoisomers are optically inactive |
| II) | $[Ma_3b_3]$ | Q) | Number of geometrical isomers = 2 |
| III) | $[Ma_3bcd]$ | R) | Number of geometrical isomers = 4 |


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| | | | |
|---|---|---|---|
| IV) | $[Ma_4bc]$ | S) | Total 3 stereoisomers |
| | | T) | Only one enantiomeric pair is possible |

A) I-P,T; II-P,R; III-R,T; IV-P,Q B) I-P,S; II-P,Q; III-R,T; IV-P,Q

C) I-P,S; II-R,Q; III-Q,T; IV-P,Q D) I-P,Q; II-P,Q; III-S,T; IV-P,Q

53. Match the following Column, Column I with Column II:

| | **Column-I** | | **Column-II** |
|---|---|---|---|
| I) | $\underline{N}(SiH_3)_3$ | P) | $p\pi - d\pi$ back bonding |
| II) | $\underline{N}(CH_3)_3$ | Q) | $sp^3$ – hybridization for underlined atom |
| III) | $\underline{B}_2H_6$ | R) | $p\pi - p\pi$ back bonding |
| IV) | $\underline{B}F_3$ | S) | Neither $p\pi - p\pi$ nor $p\pi - d\pi$ back bonding |
| | | T) | Underlined atom combines with Lewis base |

A) I-P; II-R,S; III-R,S,T; IV-R,T B) I-P; II-Q,S; III-Q,R,T; IV-S,T

C) I-P; II-Q,R; III-Q,S,T; IV-S,T D) I-P; II-Q,S; III-Q,S,T; IV-R,T

54. Match the following List-I with the List-II

| | **Column-I** | | **Column-II** |
|---|---|---|---|
| I) | PTFE | P) | Used in preparing gaskets |
| II) | Neoprene | Q) | Biodegradable |
| III) | PHBV | R) | Thermoplastic |
| IV) | PVC | S) | Produced by free radical polymerization |
| | | T) | Vulcanisation improve physical properties |

A) I-R,S; II-P,T; III-Q,T; IV-R,S B) I-P,R,S; II-P,S,T; III-Q,R; IV-R,S

C) I-R,S,T; II-P,S; III-Q; IV-P,R,T D) I-P,R; II-P,S,T; III-Q,T; IV-P,R,S



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